Data Visualization Using Tableau

Question Bank with Answers

Part A - 2 Marks

Q1. Elucidate measures in Tableau. Explain how a measure is aggregated, using the sales values ₹10,000, ₹20,000 and ₹30,000.

Introduction: A measure is a field that holds quantitative data which can be measured, aggregated and analysed mathematically. Examples are Sales, Profit, Quantity and Discount.

- **Measure:** It is a numeric field shown in the Measures section of the Data pane, normally with a green icon. Measures supply the values that are plotted on an axis, such as total sales or average profit.
- **Aggregation:** Tableau aggregates a measure when it is placed in the view. For the sales values above, `SUM(Sales)` = ₹60,000 and `AVG(Sales)` = ₹20,000. Other aggregations are MIN, MAX, COUNT, COUNTD and MEDIAN.

Q2. Articulate Measure Names and Measure Values. How are they used to compare Sales and Profit for each Category in a single view?

Introduction: Measure Names and Measure Values are two special fields generated automatically by Tableau so that several measures can be shown together in one visualization.

- **Measure Names and Measure Values:** Measure Names holds the name of each measure (Sales, Profit, Quantity), while Measure Values holds the corresponding numbers (100,000; 25,000; 250). Together they let Tableau treat many measures as one field.
- **Building the comparison view:** Place Category on Rows, Measure Values on Columns and Measure Names on Color. Tableau then draws Sales and Profit side by side for every category, with the colour legend identifying each measure.

Q3. Explain COUNT(), COUNTD() and ATTR() with an example. The Customer column contains Amit, Priya, Amit, Ravi and Priya.

Introduction: COUNT(), COUNTD() and ATTR() are aggregate functions used to count records and to display descriptive values at the current level of detail.

- **COUNT() and COUNTD():** `COUNT([Customer])` counts all non-null entries and returns 5. `COUNTD([Customer])` counts only distinct non-null values (Amit, Priya, Ravi) and returns 3. COUNTD is therefore used for unique customers or products.
- **ATTR():** It returns a value only when the field has a single distinct non-null value at the current level of detail. For a mark covering only East, `ATTR([Region])` shows East; if the mark covers East and West, Tableau displays an asterisk (*).

Q4. Elucidate the functions ISNULL(), IFNULL() and ZN() with examples. Why is NULL handling important?

Introduction: NULL represents a missing or unknown value. Comparisons involving NULL do not evaluate to TRUE or FALSE, so Tableau provides special functions to detect and replace NULL values.

- **ISNULL():** It tests whether a value is missing. `ISNULL([Sales])` returns TRUE when Sales is NULL and FALSE otherwise, so it is commonly used inside an IF condition to flag incomplete records.

- **IFNULL() and ZN():** `IFNULL([Sales], 0)` replaces a NULL with the stated alternative value. `ZN([Profit])` is the shortcut for numeric fields and returns 0 when Profit is NULL. Both prevent blanks from breaking totals.

Q5. Explain DATEDIFF() and DATEADD() with syntax and examples. An order is placed on 1 October and delivered on 6 October.

Introduction: DATEDIFF() and DATEADD() are date functions: the first measures the gap between two dates, and the second moves a date forward or backward by an interval.

```
DATEDIFF('day', [Order Date], [Delivery Date])   -- 5
DATEADD('month', 3, [Order Date])                -- 13 Jan 2027
```

- **DATEDIFF(date_part, start, end):** It returns the number of date-part boundaries between two dates. For 1 October and 6 October the result is 5 days. It is used for delivery time, customer tenure and ageing.
- **DATEADD(date_part, interval, date):** It adds or subtracts an interval. A negative interval, such as `DATEADD('day', -10, [Order Date])`, gives a date ten days earlier. It is used to compute due dates and expected delivery dates.

Q6. Write Tableau string calculations to replace "Old Model" with "Previous Model" in a product category and to find the position of the word "Science" in "Data Science". State the outputs.

Introduction: String functions manipulate, search and modify text data. Tableau counts string positions from 1, not from 0.

```
REPLACE([Product Category], "Old Model", "Previous Model")
FIND("Data Science", "Science")
```

- **REPLACE():** It substitutes one substring with another. For the input Old Model Laptop, the output is **Previous Model Laptop**. The source data is not changed; only the calculated field shows the new text.
- **FIND():** It returns the starting position of the first occurrence of the substring. For Data Science the output is **6**. If the substring does not exist in the string, FIND() returns 0.

Q7. Elucidate the number functions MOD(), DIV() and ROUND(). Find MOD(29, 6), DIV(29, 6) and ROUND(12345.678, 2).

Introduction: Number functions perform mathematical calculations on numeric fields inside calculated fields. MOD() and DIV() split a division into its remainder and its whole-number quotient.

```
MOD(29, 6)            -- 5
DIV(29, 6)            -- 4
ROUND(12345.678, 2)   -- 12345.68
```

- **MOD() and DIV():** MOD returns the remainder and DIV returns the integer quotient. Since 29 = 6 × 4 + 5, `MOD(29, 6)` = 5 and `DIV(29, 6)` = 4. Four full groups of six products can be formed, with five products left.
- **ROUND():** It rounds a number to the stated decimal places. `ROUND(12345.678, 2)` gives 12,345.68, which is useful for financial reports and dashboards that need two decimal places.

Q8. Define the Interquartile Range. Find the IQR of 5, 7, 9, 12, 15, 18 and 21 using the median-of-halves method.

Introduction: The Interquartile Range (IQR) is a measure of spread equal to the difference between the third and first quartiles: IQR = Q3 − Q1. It describes the range of the middle 50% of the data.

- **Quartiles:** The median (Q2) of the seven ordered values is 12. The lower half 5, 7, 9 has median **Q1 = 7**, and the upper half 15, 18, 21 has median **Q3 = 18**. The median itself is left out of both halves.
- **IQR and fences:** IQR = 18 − 7 = **11**. The outlier fences are 7 − 1.5(11) = −9.5 and 18 + 1.5(11) = 34.5. Every value lies inside them, so this dataset has no outlier.

Q9. Explain the Top/Bottom filter and the Context filter in Tableau with an example of each.

Introduction: Filters restrict the data displayed in a worksheet or dashboard. The Top/Bottom filter selects extreme values, while the context filter controls the order in which other filters work.

- **Top/Bottom filter:** It shows only the highest or lowest members. For example, drag Product to Filters, open the Top tab and choose Top 10 by Sales. Tableau then displays only the ten products with the highest sales.
- **Context filter:** It is applied first and creates a primary filtering condition before other filters run. It is useful for large datasets, or when one filter must limit the values offered by another, such as filtering Region before selecting the Top 5 products.

Q10. Describe how a group is created and later modified in Tableau. Use the cities Bengaluru, Chennai and Hyderabad as an example.

Introduction: A group combines several members of a dimension into one user-defined category without changing the original data source.

- **Creating a group:** Right-click the City dimension in the Data pane, choose Create → Group, select Bengaluru, Chennai and Hyderabad, click Group and name it South India. The same result is achieved by selecting two or more marks in a chart and choosing Group.
- **Modifying a group:** Right-click the group field and choose Edit Group. Members such as Kochi can be added or removed, the group can be renamed, or members can be ungrouped. All worksheets using the group update automatically.

Part B - 5 Marks

Q11. Differentiate between Groups and Sets in Tableau with a suitable example.

Introduction: Groups combine members of a dimension into broader categories, whereas Sets define a subset of data for analysis, based on selected members or on conditions.

Basis	Groups	Sets
Definition	Combines several dimension members into a single category.	Defines a subset of data based on selected members or a condition.
Purpose	To combine similar or related categories.	To identify and analyse a specific subset of data.
Based on	Usually dimension members only.	Dimensions, measures, conditions or calculations.
Dynamic nature	Generally static; changes are made by editing the group.	Can be dynamic and update automatically when the data changes.
Membership	A member is assigned to a named group category.	A member is either IN the set or OUT of the set.
Combining	Not designed for set logic.	Sets can be combined using Union, Intersection and Difference.

Basis	Groups	Sets
Use in analysis	Simplifies categories and reduces the number of members.	Compares members inside the set with members outside it.
Example	Bengaluru, Chennai and Hyderabad grouped as South India.	A Top 3 Cities by Sales set shows each city as IN or OUT.

Example: For the cities Bengaluru, Chennai, Hyderabad, Mumbai, Delhi and Kolkata, a group creates South India, North India and West India to reorganise categories. A Top 3 Cities set instead marks the three highest-selling cities as IN and the rest as OUT. Groups answer "which categories should be combined?", while sets answer "which members belong to this analytical subset?"

Q12. Differentiate between Dimensions and Measures in Tableau. Illustrate with a sales dataset.

Introduction: Dimensions are qualitative fields used to categorize, group and segment data, whereas measures are quantitative fields that are calculated, aggregated and analysed.

Basis	Dimension	Measure
Meaning	Describes or categorizes data.	Represents numerical values for analysis.
Purpose	Groups and segments data.	Calculates and summarizes data.
Typical data	Names, categories and dates.	Sales, profit and quantity.
Examples	City, Region, Product.	Sales, Profit, Quantity.
Aggregation	Usually not aggregated numerically.	Commonly aggregated using SUM, AVG, MIN and MAX.
Question answered	Who? What? Where? When?	How much? How many? What is the total or average?
Role in a chart	Defines categories or headers.	Defines values or axes.
Default colour	Blue pill (discrete).	Green pill (continuous).

Example: In records City = Bangalore, Product = Laptop, Sales = 50,000; Bangalore, Mobile, 20,000; Chennai, Laptop, 40,000; Chennai, Mobile, 30,000, City and Product are dimensions and Sales is a measure. Placing City on Rows and `SUM([Sales])` on Columns gives Bangalore ₹70,000 and Chennai ₹70,000 (grand total ₹1,40,000). Note that the colour reflects whether a field is discrete or continuous, so a dimension can be continuous and a measure can be discrete.

Q13. Differentiate between aggregate functions and table calculations. Why does Tableau report an error for SUM([Sales]) + [Profit]?

Introduction: Aggregate functions summarize many records into one value at the level of detail of the view, whereas table calculations work on the aggregated results already displayed in the view.

Basis	Aggregate function	Table calculation
Operates on	Underlying records at the level of detail of the view.	Aggregated marks already present in the view.
Examples	SUM, AVG, MIN, MAX, COUNT, COUNTD, MEDIAN.	RUNNING_SUM, RANK, quick table calculations.
Result	One summary value per mark.	A value derived from other marks, such as a cumulative total.

Basis	Aggregate function	Table calculation
Dependence on view	Depends on the dimensions in the view.	Also depends on addressing and partitioning settings.
Typical use	Total sales by region, average profit by category.	Running totals, ranking and percent of total.

```
SUM([Sales]) + [Profit]                -- error (mixed levels)
SUM([Sales]) + SUM([Profit])           -- correct
RUNNING_SUM(SUM([Sales]))             -- 1000, 3000, 6000
```

Explanation: `SUM([Sales])` is one aggregated value for a group of records, while `[Profit]` is a row-level value. Tableau cannot combine them, so both terms must be aggregated, as in the corrected formula. The running sum on the last line returns a cumulative total of 1,000, 3,000 and 6,000 across the three marks.

Q14. Throw some light on filtering data in Tableau. Explain why data is filtered and describe the main types of filters.

Introduction: Filtering means restricting the data shown in a worksheet, dashboard or visualization according to specific conditions, so that users can focus on relevant information.

- **Why filter:** Filters display only the data needed, reduce data volume, avoid overcrowded charts and let users compare particular regions, products or periods. They also improve performance and support interactive dashboards. The workflow is: select field → drag to Filters shelf → set conditions → apply.
- **Dimension filter:** It works on categorical fields such as Region or Category. Dragging Region to the Filters shelf and selecting only South displays records for the South region alone.
- **Measure filter:** It works on numeric fields. Dragging Sales to Filters and choosing At least with the value 50,000 keeps only records whose sales meet the condition.
- **Date and Top/Bottom filters:** A date filter limits the view to years, months or a date range. A Top/Bottom filter, such as Top 10 products by Sales, shows only the highest or lowest values.
- **Context and interactive filters:** A context filter is applied first and controls the values available to other filters. An interactive filter is shown on the worksheet (Region → Show Filter) so users can select North, South, East or West themselves.

Q15. Explain sorting in Tableau. Describe its types and how data is sorted by a measure.

Introduction: Sorting is the process of arranging data in a particular order so that patterns, comparisons and rankings can be understood easily. Tableau supports ascending, descending, alphabetical, manual and calculated-value sorting.

- **Ascending and descending sort:** Ascending sort arranges values from smallest to largest (10 → 50) or alphabetically (Apple → Orange). Descending sort reverses this (50 → 10 or Orange → Apple). Both can be applied from the toolbar buttons.
- **Alphabetical and manual sort:** Text dimensions can be arranged alphabetically, for example Books → Clothing → Electronics → Furniture. Manual sorting lets the user fix the order, such as North → South → East → West, when neither alphabetical nor numeric order is suitable.
- **Sorting by a measure:** A dimension can be ordered by a measure through Product → Sort → Sort By: Field → Sales → SUM → Descending. Sales of Laptop 80,000, Mobile 1,20,000, Tablet 60,000 and Printer 40,000 then appear as Mobile → Laptop → Tablet → Printer.

- **Nested sorting:** When several dimensions are present, one is sorted within another. With Region → Product → Sales, the products are arranged by sales within each region, which helps to compare performance region by region.
- **Applications and methods:** Sorting arranges bars from highest to lowest, organizes table rows, keeps line charts chronological and orders categories in dashboards. It can be applied through the sort buttons or through right-click → Sort on the dimension.

Q16. Explain the logical functions IF–THEN–ELSE, IF–ELSEIF, IIF(), CASE, and the operators AND, OR and NOT with examples.

Introduction: Logical functions evaluate conditions and return results based on whether a condition is TRUE or FALSE. They help to classify data and to apply business rules in calculated fields.

```
IF [Marks] >= 90 THEN "A+"
ELSEIF [Marks] >= 75 THEN "A"
ELSEIF [Marks] >= 60 THEN "B"
ELSEIF [Marks] >= 40 THEN "C"
ELSE "Fail"
END

CASE [Region]
WHEN "East" THEN "Zone 1"
WHEN "West" THEN "Zone 2"
ELSE "Other Zone"
END
```

- **IF–THEN–ELSE:** It tests one condition and returns one result when it is true and another when it is false. `IF [Marks] >= 40 THEN "Pass" ELSE "Fail" END` classifies every student as Pass or Fail.
- **IF–ELSEIF:** It tests several conditions in sequence and returns the result of the first true condition. In the grading code above, a score of 82 returns A, because it fails the first test but satisfies `>= 75`.
- **IIF():** It is a compact form with the syntax `IIF(condition, true_value, false_value)`. For example, `IIF([Sales] >= 10000, "High Sales", "Low Sales")` labels each sale in a single line.
- **CASE:** It compares one field against several specific values. In the code above, East returns Zone 1, West returns Zone 2 and any other region returns Other Zone. CASE is cleaner than many ELSEIF lines for exact matches.
- **AND, OR and NOT:** AND is TRUE only when both conditions are true, OR is TRUE when at least one is true, and NOT reverses the result. For example, `[Product] = "Laptop" OR [Product] = "Mobile"` selects customers who bought either product.

Q17. Explain the Average, Rank and Percentile calculations in Tableau with examples.

Introduction: Average, Rank and Percentile are calculations used to analyse sales, profit, marks and performance by finding a typical value, the position of a value and its relative standing.

- **Average:** It is the sum of all values divided by their count. For marks 60, 70, 75, 80 and 90 the average is 375 / 5 = 75. In Tableau it is written as `AVG([Marks])` or chosen as Measure → Average.
- **Rank:** It gives the position of a value among all values, and the highest value receives Rank 1. For sales of Ravi 90,000, Amit 75,000, Neha 60,000 and Priya 45,000 the ranks are 1, 2, 3 and 4, using `RANK(SUM([Sales]))`.

- **RANK versus RANK_DENSE:** For the values 100, 90, 90 and 80, standard `RANK()` returns 1, 2, 2, 4 because rank 3 is skipped. `RANK_DENSE()` returns 1, 2, 2, 3 with no gaps. Tableau also offers `RANK_PERCENTILE()` and `RANK_UNIQUE()`.
- **Percentile:** It shows the relative position of a value in the data. A student at the 90th percentile scored higher than about 90% of the observations. Tableau uses `PERCENTILE([Sales], 0.90)`, and 0.50 corresponds to the median.
- **Applications:** Average gives average order value or customer rating, rank finds the top 10 products or best regions, and percentile identifies the top 10% of customers or unusually high and low values in performance analysis.

Q18. A company has sales of ₹32,000, ₹40,000, ₹44,000, ₹56,000 and ₹68,000 for five months. Calculate the total sales, average sales, maximum sales, minimum sales and the percentage increase from the first month to the fifth month.

Introduction: These calculations use the aggregate functions SUM, AVG, MAX and MIN, together with the percentage-change formula, to summarize the monthly sales data.

- **Total sales (SUM):** ₹32,000 + ₹40,000 + ₹44,000 + ₹56,000 + ₹68,000 = **₹2,40,000**. In Tableau this is `SUM([Sales])`. It gives the total revenue earned over the five months.
- **Average sales (AVG):** Average = total ÷ number of months = ₹2,40,000 ÷ 5 = **₹48,000**. In Tableau this is `AVG([Sales])`. It represents the typical monthly sales of the company.
- **Maximum sales (MAX):** The largest of the five values is **₹68,000**, recorded in the fifth month. In Tableau this is `MAX([Sales])`. It identifies the best-performing month.
- **Minimum sales (MIN):** The smallest value is **₹32,000**, recorded in the first month. In Tableau this is `MIN([Sales])`. It identifies the weakest month and shows the starting point of the trend.
- **Percentage increase:** Increase = (Fifth month − First month) ÷ First month × 100 = (68,000 − 32,000) ÷ 32,000 × 100 = 36,000 ÷ 32,000 × 100 = **112.5%**. Sales more than doubled over the period.

Part C - 10 Marks

Q19. A company has the following quarterly sales data. Q1: Product X ₹42,000, Product Y ₹36,000, Product Z ₹28,000. Q2: X ₹46,000, Y ₹40,000, Z ₹33,000. Q3: X ₹52,000, Y ₹38,000, Z ₹37,000. Q4: X ₹60,000, Y ₹47,000, Z ₹45,000. (1) Calculate the total annual sales for each product. (2) Find the quarter with the highest total sales. (3) Calculate the percentage increase in Product X sales from Q1 to Q4. (4) Identify the most appropriate visualization (bar chart, line chart or pie chart) for the quarterly trend and justify it. (5) Explain how Measure Names and Measure Values can show all three products in one view.

Introduction: This scenario applies the SUM aggregation, percentage change and chart selection to quarterly data. Each product is a measure, and Quarter is the dimension that organizes the view.

- **Annual sales of Product X:** ₹42,000 + ₹46,000 + ₹52,000 + ₹60,000 = **₹2,00,000**. In Tableau, `SUM([Product X])` with Quarter removed from the view returns this total. It is the highest of the three products.
- **Annual sales of Product Y:** ₹36,000 + ₹40,000 + ₹38,000 + ₹47,000 = **₹1,61,000**. Product Y dipped slightly in Q3 before recovering in Q4, so its growth is less steady than that of Product X.

- **Annual sales of Product Z:** ₹28,000 + ₹33,000 + ₹37,000 + ₹45,000 = **₹1,43,000**. Product Z shows continuous growth in every quarter, although it remains the lowest in total. The grand total is ₹2,00,000 + ₹1,61,000 + ₹1,43,000 = ₹5,04,000.
- **Quarterly totals:** Q1 = 42,000 + 36,000 + 28,000 = ₹1,06,000; Q2 = ₹1,19,000; Q3 = ₹1,27,000; Q4 = ₹1,52,000. These are obtained by summing the three product measures for each quarter, or by `SUM` over Measure Values.
- **Quarter with highest sales: Q4** has the highest total of ₹1,52,000, which is about 30.2% of annual sales (152,000 ÷ 504,000). All three products reached their yearly peak in Q4, which suggests a strong year-end season.
- **Percentage increase for Product X:** Increase = (Q4 − Q1) ÷ Q1 × 100 = (60,000 − 42,000) ÷ 42,000 × 100 = 18,000 ÷ 42,000 × 100 = **42.86%**. Product X sales rose by about 42.86% between the first and last quarters.
- **Most appropriate visualization:** A **line chart** is the best choice. Quarters are an ordered time sequence, and a line connects the points so that the direction, slope and turning points of each product's trend are seen at a glance. One line per product allows direct comparison.
- **Why not bar or pie:** A pie chart shows only the share of a whole at one moment, so it cannot show change across four quarters or separate three products over time. A bar chart can compare values but hides the continuity of a trend. A line chart is therefore preferred.
- **Using Measure Names and Measure Values:** Place Quarter on Columns, Measure Values on Rows and Measure Names on Color, then filter Measure Names to Product X, Y and Z. Tableau draws one line per product in a single chart without placing each measure on the view separately.
- **Verification:** The product totals (200,000 + 161,000 + 143,000) and the quarter totals (106,000 + 119,000 + 127,000 + 152,000) both equal ₹5,04,000. Matching totals confirm that the calculations are correct, and the same check can be done in Tableau using the grand total.

Q20. A Tableau worksheet contains the following employee sales data. Deepa, East, ₹48,000, 16 orders. Farid, East, ₹60,000, 20 orders. Gopal, West, ₹36,000, 12 orders. Hema, West, ₹72,000, 24 orders. Imran, East, ₹30,000, 10 orders. Jaya, West, ₹54,000, 18 orders. Apply a Region = East filter. (1) Calculate the total sales for the East region. (2) Calculate the total number of orders for the East region. (3) Calculate the average sales per order. (4) If a sales filter of ₹45,000 or more is additionally applied, which employees remain? (5) What will be the total sales after applying both filters?

Introduction: This scenario combines a dimension filter (Region) and a measure filter (Sales). Tableau applies both filters together, so only records satisfying both conditions stay in the view.

- **Applying the Region filter:** Drag Region to the Filters shelf, select East and click OK. The view now contains only Deepa, Farid and Imran. The three West employees (Gopal, Hema and Jaya) are removed from every calculation.
- **Records that remain:** After filtering, the East records are Deepa (₹48,000, 16 orders), Farid (₹60,000, 20 orders) and Imran (₹30,000, 10 orders). All following calculations use only these three rows.
- **Total sales for East:** ₹48,000 + ₹60,000 + ₹30,000 = **₹1,38,000**. In Tableau this is `SUM([Sales])` with the Region filter applied. It is the aggregate of the filtered records only.
- **Total orders for East:** 16 + 20 + 10 = **46 orders**. In Tableau this is `SUM([Orders])`. This count is the denominator needed for the average sales per order.

- **Average sales per order:** Average = total sales ÷ total orders = 1,38,000 ÷ 46 = **₹3,000 per order**. The formula is `SUM([Sales]) / SUM([Orders])`. It is calculated from the totals, not by averaging the three employee values.
- **Applying the measure filter:** Drag Sales to the Filters shelf, choose At least and enter 45,000. The condition is checked on each remaining East record, so it works together with the Region filter and does not replace it.
- **Employees who remain:** Deepa (₹48,000) and Farid (₹60,000) satisfy both Region = East and Sales ≥ ₹45,000. **Imran** is removed because his sales of ₹30,000 are below the limit.
- **Total sales after both filters:** ₹48,000 + ₹60,000 = **₹1,08,000**. As a check, the earlier East total of ₹1,38,000 minus Imran's ₹30,000 also gives ₹1,08,000.
- **Orders and average after both filters:** The remaining orders are 16 + 20 = 36, so the average sales per order stays 1,08,000 ÷ 36 = ₹3,000. This shows that removing a low-sales record can lower the total without changing the average.
- **Combined filter logic:** Two ordinary filters behave like a logical AND. If the Region filter had to run before another filter, such as a Top N condition, it would be set as a context filter so that it is evaluated first.

Q21. The weekly sales of five stores are ₹15,000, ₹18,000, ₹18,000, ₹21,000 and ₹28,000. (1) Calculate the mean, median and mode. (2) Calculate the population and sample standard deviation. (3) Write the Tableau calculated fields. (4) Show how the mean and median change if the last store's sales are ₹98,000. (5) Interpret the results.

Introduction: Mean, median, mode and standard deviation are statistical measures that summarize data, identify typical values and measure variation. Tableau supports them through aggregate functions and frequency tables.

```
Mean Sales         : AVG([Sales])
Median Sales       : MEDIAN([Sales])
Sample SD          : STDEV([Sales])
Population SD      : STDEVP([Sales])
```

- **Mean:** Mean = sum ÷ number of observations = (15,000 + 18,000 + 18,000 + 21,000 + 28,000) ÷ 5 = 1,00,000 ÷ 5 = **₹20,000**. It is the arithmetic average and is calculated by `AVG([Sales])`.
- **Median:** Sorted values are 15,000, 18,000, 18,000, 21,000, 28,000. With five observations, the median is the value at position (5 + 1) ÷ 2 = 3, which is **₹18,000**. It is calculated by `MEDIAN([Sales])`.
- **Mode:** The value that occurs most frequently is **₹18,000**, which appears twice, whereas every other value appears once. If two values tied for the highest frequency, the dataset would have more than one mode.
- **Deviations from the mean:** The deviations are −5,000, −2,000, −2,000, +1,000 and +8,000, and they sum to 0. Their squares are 25,000,000; 4,000,000; 4,000,000; 1,000,000 and 64,000,000, giving a total of 9,80,00,000.
- **Population standard deviation:** σ = √(98,000,000 ÷ 5) = √19,600,000 ≈ **₹4,427.19**. It is used when the five stores represent the entire population of interest, and Tableau calculates it with `STDEVP([Sales])`.
- **Sample standard deviation:** s = √(98,000,000 ÷ 4) = √24,500,000 ≈ **₹4,949.75**. Dividing by n − 1 is used when the stores are only a sample of a larger population, and Tableau calculates it with `STDEV([Sales])`.

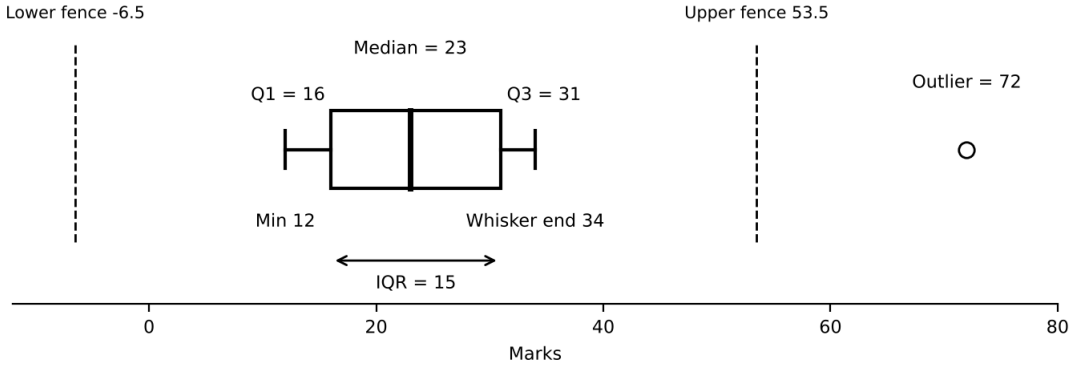
- **Tableau calculated fields:** Create the four fields shown in the code block by right-clicking in the Data pane, selecting Create Calculated Field, entering the formula and clicking OK. Dragging them into the view displays ₹20,000, ₹18,000, ₹4,949.75 and ₹4,427.19.
- **Finding the mode in Tableau:** Tableau has no built-in MODE() aggregate. Place Sales on Rows, Number of Records on Text and sort the frequency in descending order. The value with the highest frequency, ₹18,000, is the mode.
- **Effect of an extreme value:** If the last store sells ₹98,000, the mean rises to (15,000 + 18,000 + 18,000 + 21,000 + 98,000) ÷ 5 = **₹34,000**, while the median stays at **₹18,000**. Hence the median is more reliable when outliers are present.
- **Interpretation:** A standard deviation of about ₹4,427 around a mean of ₹20,000 shows moderate variation between stores. A value of zero would mean identical sales. Standard deviation measures variability only; it does not show whether performance is good or bad.

Q22. Define the Interquartile Range. For the marks 12, 14, 16, 18, 20, 23, 26, 28, 31, 34 and 72, calculate Q1, Q3, the IQR and the outlier limits, and identify any outlier. Explain how this is implemented in Tableau, including the box-and-whisker plot.

Introduction: The Interquartile Range (IQR) measures the spread of the middle 50% of the data and is calculated as Q3 − Q1. In Tableau it is used to compare variability and to detect potential outliers.

- **Definition and formula:** Q1 is the value below which about 25% of observations fall, Q2 is the median (50th percentile) and Q3 is the value below which about 75% fall. IQR = Q3 − Q1. It is less affected by extreme values than the full range.
- **Median (Q2):** The 11 values are already in ascending order. The median is the 6th value, which is **23**. In the median-of-halves method, the median is excluded from both halves.
- **First quartile (Q1):** The lower half is 12, 14, 16, 18 and 20. Its median, the third value, is **Q1 = 16**. About a quarter of the students scored 16 or less.
- **Third quartile (Q3):** The upper half is 26, 28, 31, 34 and 72. Its median, the third value, is **Q3 = 31**. About three quarters of the students scored 31 or less.
- **Interquartile range:** IQR = Q3 − Q1 = 31 − 16 = **15**. The middle 50% of the students' marks lie between 16 and 31, a spread of 15 marks.
- **Outlier fences:** Lower fence = Q1 − 1.5 × IQR = 16 − 22.5 = **−6.5**. Upper fence = Q3 + 1.5 × IQR = 31 + 22.5 = **53.5**. Any value outside these limits is a potential outlier.
- **Identifying the outlier:** The mark **72** is greater than 53.5, so it is an outlier. All other marks, from 12 to 34, lie between −6.5 and 53.5 and are normal. The highest non-outlier value, 34, becomes the end of the upper whisker.
- **Tableau calculated fields:** Create Q1 as `PERCENTILE([Marks], 0.25)`, Q3 as `PERCENTILE([Marks], 0.75)` and the IQR as their difference. An Outlier Status field then uses IF…OR…THEN “Outlier” ELSE “Normal” with the two fences, as in the code block below.
- **Box-and-whisker plot:** Drag Department to Columns and Marks to Rows, open the Analytics pane and drag Box Plot into the view. Tableau draws the box from Q1 to Q3, the median line, whiskers and outlier points, as in the figure below, which makes departments easy to compare.
- **Cautions and applications:** Quartile values vary slightly by software method, and PERCENTILE() depends on data-source support, so an LOD expression or table calculation may be needed. IQR is used in student performance, sales analysis, finance, healthcare and quality control.

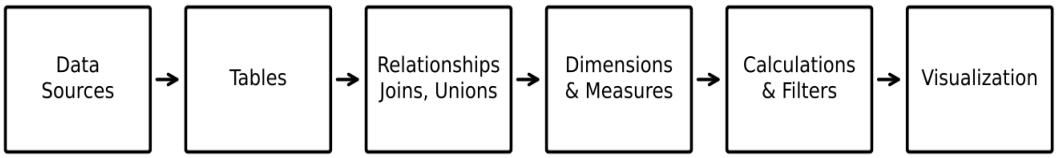
```
Q1                : PERCENTILE([Marks], 0.25)
Q3                : PERCENTILE([Marks], 0.75)
IQR               : [Q3] - [Q1]
Outlier Status    : IF [Marks] < [Q1] - 1.5 * [IQR]
                    OR [Marks] > [Q3] + 1.5 * [IQR]
                    THEN "Outlier" ELSE "Normal" END
```



Box-and-whisker plot for the marks data (IQR = 15, outlier = 72)

Q23. Explain the Data Layer in Tableau with a neat diagram. Discuss its main components and how they work together.

Introduction: The Data Layer is the underlying structure and organization of data that Tableau uses to connect, combine, prepare and analyse information. It forms the foundation for every visualization, dashboard and report.



Flow of the Data Layer in Tableau

- **Data sources:** Tableau connects to databases, spreadsheets, cloud services and files. Examples are Excel, CSV, MySQL, SQL Server, Oracle and Google Sheets. The data source is the starting point of the flow, and every later step depends on it.
- **Tables:** Tables hold the actual data organized into rows and columns, such as customer, sales, product, employee or transaction information. Each column becomes a field in the Data pane.
- **Relationships:** Relationships connect tables through a common field. For example, Customer ID connects a Customer table with an Orders table, so that data from both can be analysed without physically merging them.
- **Joins:** Joins combine tables using a common field. The Inner Join keeps only matching rows, the Left Join keeps all rows of the left table, the Right Join keeps all rows of the right table and the Full Outer Join keeps all rows of both.
- **Unions:** A union combines rows from tables that have a similar structure. For example, the January Sales and February Sales tables can be stacked into a single table so that both months are analysed together.
- **Dimensions and measures:** Dimensions are categorical fields such as Product, Region and Customer that describe or group data. Measures are numerical fields such as Sales, Profit and Quantity that can be aggregated.

- **Data types:** Tableau identifies each field as text, number, date or Boolean. Correct data types are important, because date functions need date fields and aggregations such as SUM need numeric fields.
- **Calculated fields:** Users create new fields from formulas. For example, Profit Ratio = `SUM([Profit]) / SUM([Sales])`. Calculated fields extend the data layer without altering the source.
- **Filters:** Filters restrict the data used in analysis, for example showing only 2026 sales or one region. They reduce data volume and let dashboards focus on what users need.
- **Worked example:** For records Customer 101, Laptop, North, ₹50,000, ₹8,000 profit; 102, Mobile, South, ₹30,000, ₹5,000; and 103, Laptop, West, ₹45,000, ₹7,000, the data layer supports Sales by Region, Profit by Product, Top Customers and Profit Margin.

Q24. A retail company's worksheet has the fields [Customer Name], [Product Code], [Order Date], [Delivery Date] and [Sales]. Write calculated fields and explain the output for the sample record Customer Name " rahul sharma ", Product Code ELE1025, Order Date 13 October 2026, Delivery Date 18 October 2026 and Sales ₹65,000, to (1) show the cleaned name in uppercase, (2) extract the surname, (3) extract the category code, (4) find the delivery days, (5) find the expected delivery date if delivery takes 14 days, (6) show the month name, (7) classify the sales, (8) handle NULL profit, (9) find the percentage contribution of a product with sales ₹20,000 out of ₹80,000, and (10) rank salespeople by sales.

Introduction: Calculated fields combine string, date, logical, number and aggregate functions to clean data and derive new information. Each formula is created through Create Calculated Field in the Data pane.

```
1. UPPER(TRIM([Customer Name]))
2. PROPER(SPLIT(TRIM([Customer Name]), " ", 2))
3. LEFT([Product Code], 3)
4. DATEDIFF('day', [Order Date], [Delivery Date])
5. DATEADD('day', 14, [Order Date])
6. DATENAME('month', [Order Date])
7. IF [Sales] >= 50000 THEN "Excellent"
   ELSEIF [Sales] >= 25000 THEN "Good"
   ELSE "Needs Improvement" END
8. ZN([Profit])
9. ROUND(([Product Sales] / [Total Sales]) * 100, 2)
10. RANK(SUM([Sales]))
```

- **Cleaned name (1):** TRIM removes the leading and trailing spaces and UPPER converts the text to capitals, so " rahul sharma " becomes **RAHUL SHARMA**. Nesting functions standardizes imported text for reports and comparisons.
- **Surname (2):** After TRIM, `SPLIT(…, " ", 2)` returns the second word, sharma, and PROPER capitalizes it to **Sharma**. This assumes that the surname is the second word; names with extra words need a different formula.
- **Category code (3):** `LEFT([Product Code], 3)` extracts the first three characters, so ELE1025 gives **ELE**. Such codes identify the product category, and the field can then be used for grouping.
- **Delivery days (4):** `DATEDIFF('day', …)` counts the day boundaries between 13 October and 18 October and returns **5 days**. The field can be averaged by region to study delivery performance.
- **Expected delivery date (5):** `DATEADD('day', 14, [Order Date])` adds 14 days to 13 October 2026 and gives **27 October 2026**. It is useful for promised delivery dates and deadlines.

- **Month name (6):** `DATENAME('month', …)` returns the month as text, so the order date gives **October**. Unlike DATEPART, which returns the number 10, the name is suitable for labels and month-wise reports.
- **Sales classification (7):** Tableau tests the conditions in order. As ₹65,000 satisfies `>= 50000`, the result is **Excellent**. A sale of ₹35,000 would return Good, and ₹15,000 would return Needs Improvement.
- **NULL handling (8):** `ZN([Profit])` returns 0 when profit is NULL and the original number otherwise. This prevents blank values from distorting totals, averages and later calculations.
- **Percentage contribution (9):** (20,000 ÷ 80,000) × 100 = **25%**, and ROUND keeps two decimal places. The total must be computed at the correct level of aggregation, otherwise the percentage will be wrong.
- **Rank (10):** `RANK(SUM([Sales]))` is a table calculation that gives Rank 1 to the highest total. Ravi with ₹90,000, ahead of Amit with ₹75,000, would receive rank 1 and Amit rank 2, which supports top-performer analysis.